

Assume G is in Chomsky normal form.

Example. $S \rightarrow AB \mid BA \mid SS \mid AC \mid BD$

$A \rightarrow a, B \rightarrow b, C \rightarrow SB, D \rightarrow SA$

$L(G)$ - all strings with equal number of a 's and b 's.

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Let $x = aab bab$. Let $n = |x|$ (So $n=6$ here)

	a	a	b	b	a	b
0	1	2	3	4	5	6

For $0 \leq i < j \leq n$, let x_{ij} denote the substring of x between i & j . Ex. $x_{0,3} = aab$, $x_{2,6} = bbab$
 $x = x_{0,n}$.

0						
-	1					
-	-	2				
-	-	-	3			
-	-	-	-	4		
-	-	-	-	-	5	
-	-	-	-	-	-	6

Entry T_{ij} of T - Set of non-terminals of G that generates the substring x_{ij} of x .

Define by induction on the length of the substrings.

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$x \rightarrow$ a a b b a b
0 1 2 3 4 5 6

Substring of length 1.

0						
<u>A</u>	1					
—	<u>A</u>	2				
—	—	<u>B</u>	3			
—	—	—	<u>B</u>	4		
—	—	—	—	<u>A</u>	5	
—	—	—	—	—	<u>B</u>	6

Substring of length ≥ 3

0						
<u>A</u>	1					
<u>∅</u>	<u>A</u>	2				
<u>∅</u>	<u>S</u>	<u>B</u>	3			
<u>S</u>	<u>C</u>	<u>∅</u>	<u>B</u>	4		
<u>D</u>	<u>S</u>	<u>∅</u>	<u>S</u>	<u>A</u>	5	
<u>S</u>	<u>C</u>	<u>∅</u>	<u>C</u>	<u>S</u>	<u>B</u>	6

For each $x_{i,i+2}$,
break into 2 substring
 $x_{i,i+1}$ & $x_{i+1,i+2}$

Select a nonterminal
 X from $T_{i,i+1}$ and
 Y from $T_{i+1,i+2}$

Look for production
 $Z \rightarrow XY$ in G .

Label $T_{i,i+2}$ with Z
for each such production.

$S \in T_{0,6}$ indicates $S \xrightarrow{*}_G x_{0,6} = x$: the input string.
Therefore $x \in L(G)$.

Algorithm

```
For  $i := 0$  to  $n-1$  do
  begin
     $T_{i,i+1} = \phi$ 
    For  $A \rightarrow a$ , a production in  $G$  do
      if  $a = x_{i,i+1}$  then  $T_{i,i+1} = T_{i,i+1} \cup \{A\}$ 
  end
```

[length 1 strings].
[Initialise to ϕ]

```
For  $m := 2$  to  $n$  do
  For  $i := 0$  to  $n-m$  do
    begin
       $T_{i,i+m} := \phi$ 
      For  $j = i+1$  to  $i+m-1$  do
        For  $A \rightarrow BC$ , a production in  $G$  do
          if  $B \in T_{i,j}$  and  $C \in T_{j,i+m}$ 
            then
               $T_{i,i+m} = T_{i,i+m} \cup \{A\}$ 
      end
    end
```

[length $m \geq 2$]
[for each substring of length m]
[Initialise]
[All possible splits]

Running Time - $O(n^3 p)$ where $p = |P|$ and $n = |x|$.

Ambiguous Grammar.

Consider the grammar $S \rightarrow S+S \mid S*S \mid (S) \mid A$
 $A \rightarrow a \mid b$

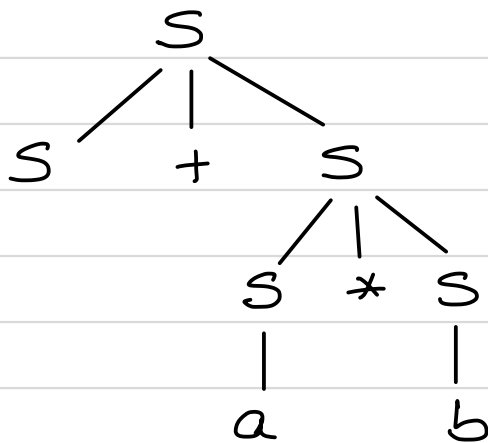
Consider the string $a+a*b$

Derivation 1. $S \rightarrow S+S \rightarrow a+S \rightarrow a+S*S \rightarrow a+a*b$
 $S \rightarrow S+S \rightarrow S+S*S \rightarrow$

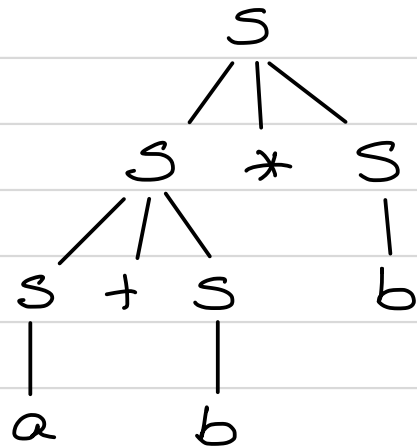
Derivation 2. $S \rightarrow S*S \rightarrow S+S*S \rightarrow a+S*S \rightarrow a+a*b$

Parse Tree

Derivation 1



Derivation 2.



A CFG G is ambiguous if $\exists x \in L(G)$ for which there are two different parse trees.

↳ Not two different derivations.

Definition

A string α is derived ambiguously in a CFG G if it has two different leftmost derivation.

Grammar G is ambiguous if it generates some string ambiguously.

G is unambiguous if G is not ambiguous.

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A CFL $A \subseteq \Sigma^*$ is inherently ambiguous if $\forall$ CFG G s.t. $L(G) = A$, G is ambiguous.

Note. There are inherently ambiguous CFLs.